

DS-UA 202: Responsible Data Science

Course Project Report

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Project Overview

ADS analyzed. We audit an algorithmic decision-support system (ADS) used by a music streaming platform (“Playlist Pro”) to predict customer churn and target retention discounts. The model is a PyTorch multi-layer perceptron (MLP) trained on demographic and behavioral features such as age, location, weekly listening hours, and customer-service inquiries.

Why this ADS matters. The output drives a direct financial intervention: users predicted to churn are offered discounts, while users predicted not to churn are not contacted. Because this allocates a monetary benefit, fairness and subgroup performance are central concerns rather than secondary diagnostics.

Sources.

- Kaggle Competition: <https://www.kaggle.com/competitions/streaming-subscription-churn-model/leaderboard>
- Baseline implementation: <https://www.kaggle.com/code/ethanjychen/streaming-churn-rate>

1 Background: General information about the ADS

a. Purpose and stated goals

The purpose of this ADS is to identify users likely to cancel their subscriptions so the platform can intervene early with retention offers. Its stated business goal is to reduce churn efficiently by sending discounts to users with high predicted churn risk, instead of offering blanket discounts to everyone.

b. Multiple goals and trade-offs

The ADS has multiple goals that create explicit trade-offs:

- **Retention effectiveness:** maximize recall to catch as many churners as possible.
- **Budget efficiency:** maximize precision to avoid unnecessary discounts.

- **Fair access to intervention:** avoid concentrating false negatives in specific demographic or subscription groups.

In this ADS, false negatives are particularly costly because they are at-risk users who do not receive intervention. Therefore, maximizing overall accuracy alone is insufficient if the error burden is unevenly distributed.

2 Input and output

a. Data used and data collection/selection

We use the Kaggle “Streaming Subscription Churn Model” dataset (125,000 users, 19 input features). The dataset is synthetic but designed to resemble records that a streaming platform would collect from activity logs, billing systems, and customer support systems. Data is prepackaged by the competition and selected for supervised churn prediction.

b. Input features: datatype, missingness, distribution, profiling

The dataset includes both numeric and categorical attributes. Missingness is negligible (effectively none in the provided synthetic files). Table 1 profiles representative core features used in the audit.

Table 1: Representative Input Feature Profiling

Feature	Datatype	Missing	Distribution/Notes
customer_id	Integer	None	Unique identifier; dropped for modeling.
age	Integer	None	Broad range (~18–70+); used for generation subgrouping.
location	Categorical	None	Multi-class geographic region/state variable; roughly balanced categories.
subscription_type	Categorical	None	Free/Student/Family/Premium; key fairness grouping variable.
payment_plan	Categorical	None	Monthly/yearly style categories.
weekly_hours	Float	None	Right-skewed engagement feature; high-usage tail.
customer_service_inquiries	Integer	None	Count-valued variable; higher values associated with churn.
num_subscription_pauses	Integer	None	Non-negative count feature; moderate predictive influence.

Additional behavioral and transactional features (for example, skip rate, notifications clicked, social engagement features, payment method, and signup-related fields) complete the 19-feature set.

Pairwise correlation and other profiling. Pairwise correlation checks on numeric variables indicate that no single pair dominates with extreme collinearity, but meaningful patterns exist: `customer_service_inquiries` is positively associated with churn-like behavior, while `weekly_hours` is negatively associated. This is important for fairness because behavioral variables can proxy demographic or socioeconomic attributes.

c. System output and interpretation

The model output is a **probability score** in $[0, 1]$ from a sigmoid layer, interpreted as predicted churn risk. The implementation converts this score into a **binary class label** using a threshold (0.5 in the baseline):

- 1 (predicted churn): user is targeted for a discount intervention.
- 0 (predicted retain): user receives no intervention.

Thus, model predictions directly determine allocation of financial retention resources.

To objectively measure these disparities, we established strict fairness benchmarks. We applied the EEOC’s 80% Rule to evaluate Selection Rate, ensuring the retention budget is distributed equitably. More importantly, we optimized for the False Negative Rate and set a strict 5% gap threshold. In this context, a false negative is not just a classification error—it is a “missed churning,” an at-risk customer who is structurally denied a financial discount they deserve.

3 Implementation and validation

a. Data cleaning and preprocessing

The baseline notebook performs standard preprocessing:

- encodes categorical features,
- scales numeric features,
- removes non-predictive identifiers (such as `customer_id`),
- splits data into training and held-out validation subsets.

These steps are consistent with common tabular deep-learning pipelines.

b. High-level implementation details

The ADS uses a PyTorch MLP (feedforward neural network) trained with binary cross-entropy loss and Adam optimization. At inference, each user vector is passed through hidden layers and a sigmoid output node. The resulting risk score is thresholded to produce the churn/no-churn label used by the intervention system.

c. Validation and whether goals are met

We validate on a held-out 20% split ($\sim 25,000$ users) using Accuracy, Precision, Recall, F1-score, and AUC-ROC.

These results indicate strong aggregate predictive performance and suggest that the model meets its narrow technical objective (ranking churn risk reasonably well). However, whether it meets the broader responsible-deployment objective requires subgroup and fairness analysis, which we report next.

Table 2: Overall Validation Metrics

Metric	Value
Accuracy	0.841
Precision	0.822
Recall	0.880
F1-score	0.850
AUC-ROC	0.932

4 Outcomes

a. Accuracy across subpopulations and metric justification

We prioritize recall because the intervention is benefit allocation: a false negative is a missed at-risk user who does not receive a potentially churn-preventing offer. We also report precision, FPR, and FNR to characterize both budget efficiency and harm distribution.

Table 3: Accuracy Metrics by Generation

Generation	Size	Precision	Recall	FPR	FNR
Gen Z	4,118	0.828	0.916	0.267	0.084
Millennials	6,509	0.800	0.861	0.179	0.139
Gen X	6,328	0.785	0.847	0.160	0.153
Boomers	8,045	0.853	0.892	0.239	0.108

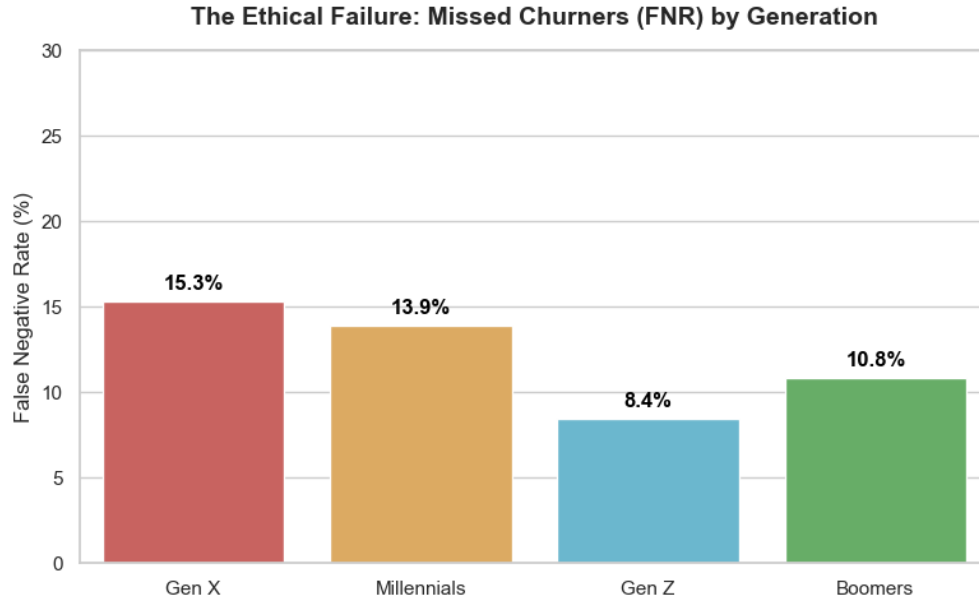


Figure 1: False negative rate by generation from the updated notebook output.

The largest disparity is across subscription tiers: recall is 96.4% for Free users but 76.1% for

Table 4: Accuracy Metrics by Subscription Type

Subscription	Size	Precision	Recall	FPR	FNR
Free	6,173	0.889	0.964	0.484	0.036
Student	6,303	0.805	0.892	0.296	0.108
Family	6,182	0.757	0.785	0.130	0.215
Premium	6,342	0.754	0.761	0.127	0.239

Premium users. This means paid-tier churners are disproportionately missed by the model.

b. Fairness analysis and metric justification

We use three fairness metrics because each captures a different normative concern.

Equalized odds (error-rate parity). We focus on FNR parity since false negatives correspond to denied intervention. FNR varies from 3.6% (Free) to 23.9% (Premium), showing substantial unequal treatment in misses.

Demographic parity (selection-rate parity). Selection rates are much higher for Free users than paid tiers, indicating unequal access to discounts.

Predictive parity (precision parity). Precision differs by group (e.g., 0.889 Free vs. 0.754 Premium), meaning reliability of churn alerts is not uniform.

Intersectional analysis amplifies this picture:

- best subgroup FNR: 0.017 (Gen Z – Free),
- worst subgroup FNR: 0.304 (Millennials – Premium),
- disparity: $> 17\times$.

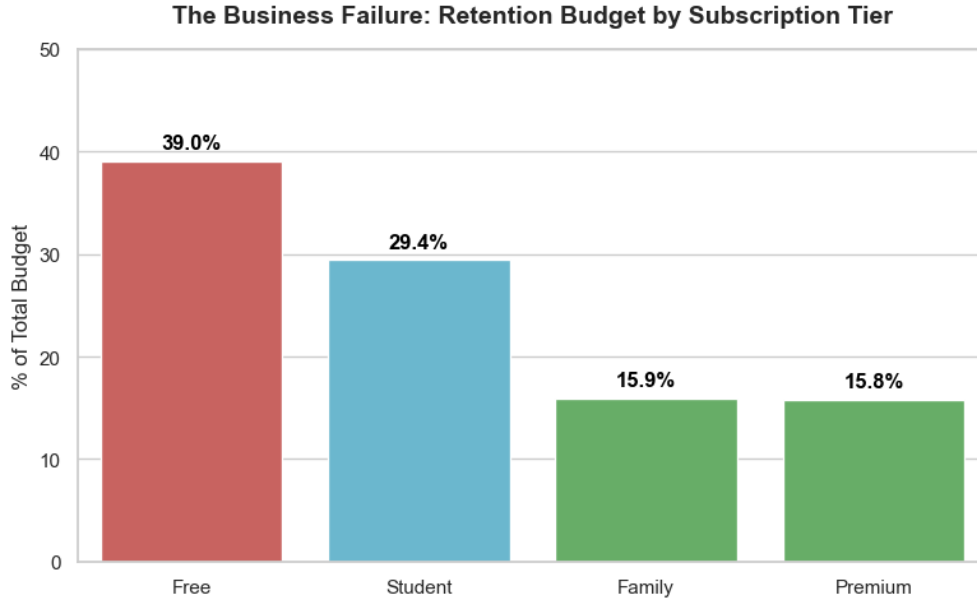


Figure 2: Retention budget allocation by subscription tier from the notebook output.

c. Additional audit methods and justification

Beyond point-metric fairness, we audit interpretability, robustness, and downstream allocation impact.

Interpretability. The notebook diagnostics indicate that behavioral engagement and service-contact signals drive much of model behavior, while subscription-related variables remain highly influential. This supports the proxy-risk interpretation used in our fairness discussion.

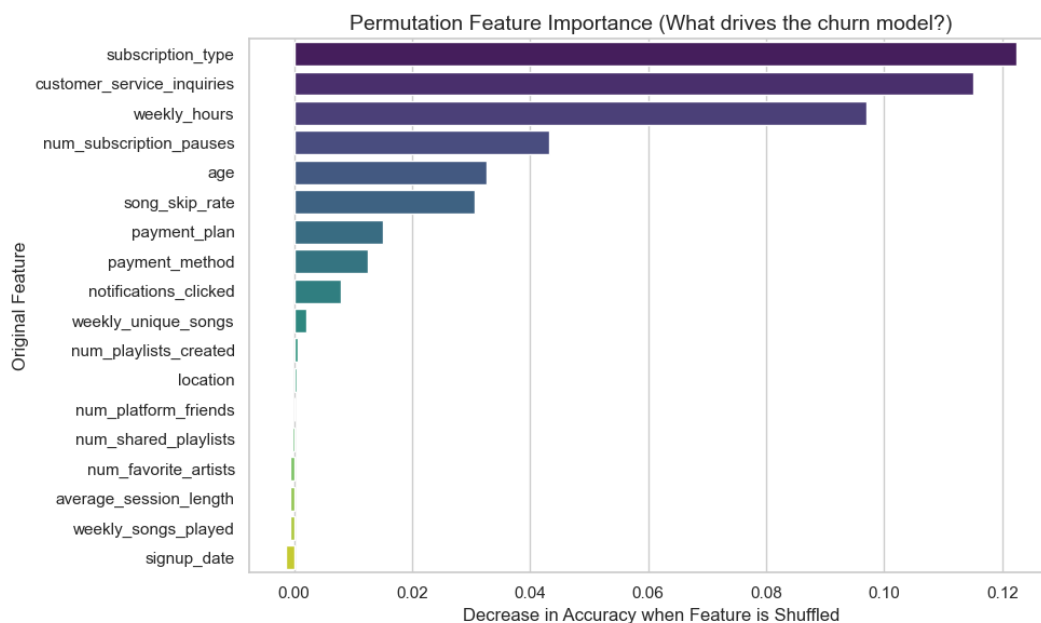


Figure 3: Permutation feature importance used in the additional audit methods.

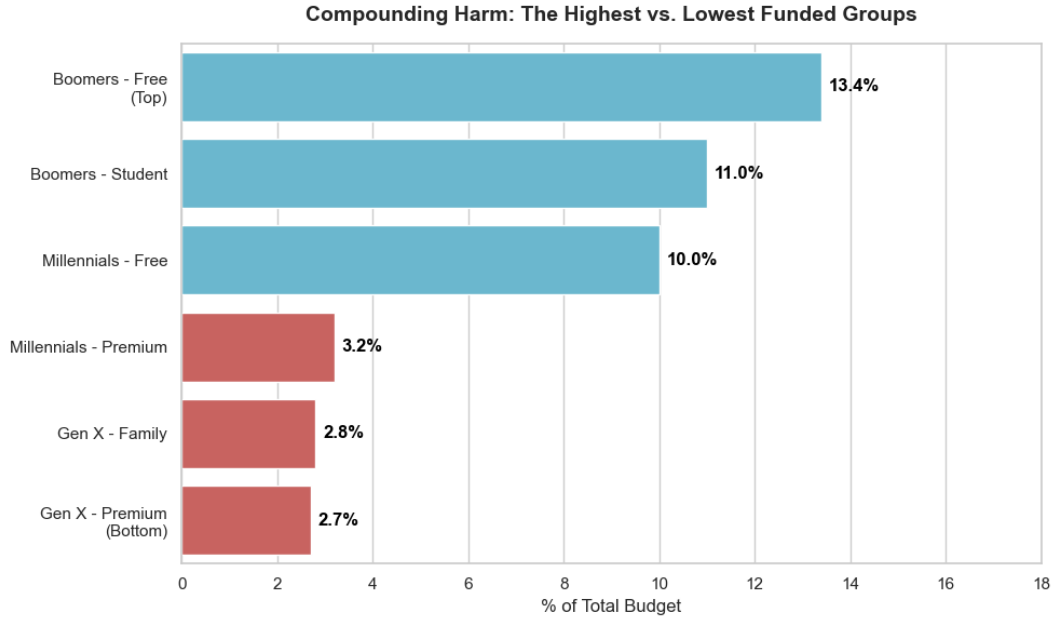


Figure 4: Intersectional harm visualization from the notebook output.

Downstream impact (budget allocation).

Table 5: Downstream Budget Allocation by Age Bracket

Age Bracket	Discounts Given	Budget (\$)	% of Total
18–25	2,179	10,895	15.9%
26–35	2,152	10,760	15.7%
36–50	2,694	13,470	19.6%
51+	6,722	33,610	48.9%

Table 6: Downstream Budget Allocation by Subscription Type

Subscription Type	Discounts Given	Budget (\$)	% of Total
Family	2,181	10,905	15.9%
Free	5,360	26,800	39.0%
Premium	2,166	10,830	15.8%
Student	4,040	20,200	29.4%

This method is appropriate because it translates model behavior into resource-allocation consequences, which is the real policy object of the ADS.

5 Summary

a. Was the data appropriate for this ADS?

Partially. The dataset is large and feature-rich enough for churn modeling, but synthetic-generation assumptions, static snapshots, and missing direct socioeconomic features limit confidence in real-world generalization and fairness interpretation.

b. Is the implementation robust, accurate, and fair? Which stakeholders may prefer these metrics?

The implementation is accurate in aggregate but not robustly fair across groups. Accuracy metrics (AUC/recall/precision) are useful for business optimization, while FNR parity and selection-rate parity are more appropriate for consumer-protection and governance stakeholders. Business teams may prioritize recall for high-value users; users and ethicists may prioritize equalized-odds-style protections against unequal miss rates.

c. Would we be comfortable deploying in public sector or industry?

No. In industry, the model under-identifies churn among paid tiers and misaligns intervention spending. In public-sector-like deployments, proxy discrimination concerns and unequal access to benefits create unacceptable fairness and accountability risks.

d. Recommended improvements to data, processing, or analysis

1. Collect longitudinal engagement trajectories rather than single snapshots.
2. Improve data provenance and document synthetic-generation assumptions clearly.
3. Reduce proxy dependence on subscription-tier variables through fairness-aware modeling.
4. Apply subgroup-aware calibration and monitor FNR parity during pilot deployment.
5. Add regular re-auditing with intersectional fairness dashboards.

6 Project contributions

Trem Ampeloquio

- Led fairness-metric design and subgroup evaluation (generation, subscription, and intersectional analyses).
- Wrote substantial portions of the outcomes and fairness discussion sections.
- Interpreted downstream impact and deployment implications.

Terry Han

- Led implementation review of the baseline Kaggle/PyTorch pipeline and preprocessing flow.

- Produced core validation outputs and supported robustness checks.
- Contributed to summary recommendations and final report editing.

Shared work

- Jointly selected the ADS, scoped audit questions, and validated final narrative coherence.
- Jointly reviewed all tables and results for consistency with responsible-data-science framing.